What a Tooth-Level Score Cannot Tell You: Constant Baselines, External Validation and Domain Shift in Automated Panoramic Dental Screening

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***Abstract*—Tooth decay (dental caries) and gum disease that destroys the bone around teeth (periodontal bone loss) are routinely screened for on panoramic dental radiographs, and a growing number of artificial-intelligence systems are reported to detect them automatically. Such systems are almost always scored by asking which teeth they mark as diseased. We show that this score is difficult to interpret, because decay is concentrated in the back teeth: a program that ignores the radiograph entirely and always names the same eight molars can score well. We built a complete six-stage pipeline that finds each tooth, numbers it, and classifies it for decay and for bone loss, and we compared it against progressively less informed systems—two general-purpose vision–language models and, finally, that image-blind constant—using one scorer on one set of 164 radiographs. The constant scored highest (F1 0.320 versus our pipeline's 0.294), and a second metric that requires the flagged tooth to sit where the lesion actually is produced the same ordering (0.488 versus 0.443). On 337 radiographs from four hospitals that contributed no training data, the comparison produced no separation at all, and the difference between the two systems swung 26 percentage points across sites; on a second, larger external benchmark of 3,421 lesions the constant won outright, by 25 percentage points. Both language models returned confident, correctly formatted tooth lists for an image containing pure random noise. The bone-loss branch scored 0.923 macro-F1 on the data it was developed on and 1.4–5.6 % sensitivity on the external hospitals. We do not conclude that automated caries screening fails. We conclude that a tooth-level score alone cannot establish that a system is reading the radiograph, and that four things a constant cannot do—identifying which tooth, reporting a confidence that behaves like a probability, declining when uncertain, and being measured on outside institutions—are where the evidence for a working system has to come from.**

***Index Terms*—panoramic radiography, dental caries, periodontal bone loss, evaluation methodology, image-blind baselines, external validation, model calibration, selective prediction, domain shift**

I. Introduction

The clinical setting

A panoramic dental radiograph—an *orthopantomogram*, or OPG—is a single X-ray image that captures the whole upper and lower jaw in one exposure. It is the most common extraoral radiograph taken in dentistry, because one image at a low radiation dose shows every tooth, the bone that supports them, and the jaw joints. Dentists use it to survey a mouth quickly: to find teeth that need treatment, to plan extractions and implants, and to check for disease a patient has not yet noticed.

Two diseases dominate that survey. **Dental caries** is tooth decay: acid produced by bacteria dissolves the hard tissue of the tooth, first the enamel surface and then the softer dentine beneath. On a radiograph it appears as a dark (radiolucent) area where solid tooth should be. Caught early, it is treated with a filling; missed, it reaches the nerve and requires root-canal treatment or extraction. **Periodontal disease** is inflammation of the tissues that hold a tooth in place. In its advanced form it destroys the surrounding bone, and that destruction—**periodontal bone loss**—is what a radiograph can show: the bone level around the tooth root sits lower than it should. It is the leading cause of tooth loss in adults, and it is painless until late.

Both are common, both are visible on an OPG, and both are missed. Panoramic images are geometrically distorted, teeth overlap one another, and a busy clinician reads many of them quickly. Automated screening is attractive for exactly this reason: not to replace the dentist's judgement, but to make a second pass over an image a human has already read, and flag what may have been overlooked.

What automated screening currently reports

Deep learning on OPGs has converged on a standard decomposition: detect the individual teeth, then classify each one. Systems built this way report caries and periodontal findings at accuracies that read as clinically useful [1]–[3]. The result is almost always reported as a **tooth-level** score: of the teeth in this mouth, which ones does the system say are diseased, and how well does that list agree with the expert's.

The evaluation problem, by example

Consider a program that contains no neural network, no image processing, and no pixels—only a fixed list of eight tooth codes: the first and second molars in all four corners of the mouth. It emits that same list for every radiograph it is ever shown. It cannot see.

On our development data, that program achieves an F1 score of **0.320** on the tooth-level caries question. Our six-stage pipeline, which does read the image, achieves **0.294**.

The reason is not that the pipeline is broken. It is that caries is not spread evenly around the mouth: molars have deep grooves, are hardest to clean, and carry most of the disease. A tooth-level score asks which members of a fixed set of 32 tooth positions are affected—a question that population statistics answer partially correctly without reference to any particular patient. A fixed answer therefore starts with a substantial score already in hand, and the room left above it is small.

Why that matters

This is a statement about *evidence*, not about anyone's model. A published tooth-level score is not wrong. But on its own it cannot distinguish a system that recognises decay in this patient's radiograph from a system that has learned where decay usually is. The two produce similar numbers and very different clinical objects: only the first can find the lesion in the place the population statistics do not expect, which is precisely the lesion a hurried human reader also missed.

The natural response is to check what the floor is. We reviewed 49 task-relevant papers drawn from our own survey of this area. **Two** report validation on data from an institution other than the one that produced their training set. We found no report of an image-blind or otherwise uninformed baseline in any of them.

What we did

We measured that floor on our own work rather than anyone else's. We first built a system worth testing: a six-stage pipeline performing tooth detection, tooth numbering, caries classification and periodontal bone-loss classification, with confidence intervals, calibrated probabilities, a threshold for declining uncertain cases fixed before the data were seen, and evaluation on four hospitals that contributed no training data. We then constructed a ladder of opponents, each given strictly less information than the one above, and scored every rung with one scorer on one set of images.

Testing on outside institutions matters for a specific reason. A model learns the scanner, the exposure settings, the patient population and the annotation habits of the place its training data came from, along with the disease. Only data from somewhere else can separate the two. We report every external result per hospital, because pooling four sites into one number conceals exactly the variation the exercise exists to expose.

Contributions

1) An **image-blind constant-baseline protocol** for tooth-level OPG evaluation, fitted without leakage and applied across two unrelated metric families and six independent cohorts.
2) A **vision–language-model baseline with the controls that make it interpretable**, including what these models answer when shown an image containing no anatomy at all.
3) **External validation on four hospitals and a second public benchmark**, reported per site, with the spread that pooling conceals.
4) **Calibration and selective-prediction** results, each with a control that isolates the mechanism responsible, and each reported with its cost rather than only its gain.
5) An honest characterisation of the pipeline, including a periodontal branch that does not survive domain shift and a headline number that is bounded by an earlier stage.

We are explicit about what we do not claim. We do not assert that published caries results are incorrect, and we do not claim our own system is better than any of them. We argue something narrower: that a tooth-level number reported without a floor cannot be distinguished from an anatomical prior.

II. Related Work

A. Detection and classification on panoramic radiographs

Detector choice for this data is well studied. A head-to-head evaluation on dental panoramics reports YOLOv11 outperforming its successor generation for multi-class dental detection, and observes an asymmetry that matters here: performance is strong on well-defined objects such as crowns and implants and markedly weaker on subtle radiolucent pathology, including caries and bone loss [4]. YOLO-family detectors are widely applied to OPGs more generally [5]–[7]. Caries and periodontal classification have been approached with dual-network frameworks [1], dedicated convolutional classifiers [3], and contrastive feature-pyramid designs [2], alongside broader detection and segmentation work [8], [9].

B. Tooth numbering, and two different tasks with one name

Tooth numbering is typically reported at very high accuracy—one recent system reports 0.9988 [10], and multi-stage segmentation-and-numbering pipelines report comparable figures [11]. Such numbers are frequently obtained on *pre-cropped* teeth, where the task is a 32-way classification of an image already known to contain exactly one tooth, correctly framed. That is a materially easier problem than numbering under detector output, where teeth may be missed, duplicated or clipped. We report both framings and find the gap between them large.

C. Vision–language models in dentistry

A **vision–language model** (VLM) is a general-purpose model that takes an image and a written instruction and replies in text. Recent work applies general-purpose and domain-adapted VLMs to dental imaging [12]–[15]. They are attractive because they need no task-specific training, and difficult to evaluate for the same reason: a fluent, confident, correctly formatted answer is not evidence that the model perceived anything. We adopt a control from the shortcut-learning literature [16]—present an input containing none of the signal the model purports to read, and observe whether its behaviour changes.

D. External validation and the missing floor

External validation remains uncommon in this subfield. In our review of 49 task-relevant papers, two evaluate on an outside institution. Where it is done, degradation is often severe: one dental object-recognition system falls from 95.06 mAP on its own data to 61.83 on a public benchmark [17]. Independent work on periodontal bone loss, evaluated against five clinicians, reports weighted F1 of 0.29 with severe-class F1 of zero [18]—a reminder that in-domain figures can be a poor guide to behaviour elsewhere. To our knowledge no work in this area reports an image-blind baseline, which is the gap this paper addresses.

III. System

The finding in this paper is only interesting if the system it is measured against is a serious one. We therefore describe the pipeline before the evaluation that tests it.

A. Architecture

The pipeline maps an OPG to a structured per-tooth record in six stages (Table I).

Stage 1 detects individual teeth with a YOLOv11l segmentation model [19], emitting one box per tooth.

Stage 2 assigns each detected tooth its clinical name. We use **FDI two-digit notation**, the international standard: the first digit is the quadrant of the mouth (1 = upper right, 2 = upper left, 3 = lower left, 4 = lower right) and the second is the position counting outwards from the midline (1 = central incisor, through to 8 = third molar, the wisdom tooth). Tooth 16 is therefore the upper-right first molar and tooth 47 the lower-right second molar. Numbering matters because everything downstream is reported against it: a caries verdict on the wrong tooth code is a wrong clinical finding, however good the classifier. A ResNet18 [20] head consumes the tooth crop together with four positional features, and the resulting scores are resolved to a one-to-one assignment across the mouth by the Hungarian algorithm [21], so no two teeth receive the same code.

Stage 3 classifies each tooth for caries and **Stage 4** for periodontal bone loss, using independent ResNet18 branches. Each branch is a coarse-to-fine cascade: a binary gate decides whether the finding is present, and a second model grades severity only on teeth the gate flags.

Stages 5 and 6 fuse the per-stage outputs on the FDI key and emit the per-tooth record.

The complete system runs at **2.39 s per image on CPU** with no GPU present. Both figures become relevant in Section VI.

B. Two diseases, one of which is reported as a negative result

The periodontal branch is a full component of the implemented system, and its result is a measured failure. We keep it in the paper for that reason rather than in spite of it: it is the clearest demonstration available to us that an in-domain score on a periodontal task is not evidence of a usable periodontal model. It is reported throughout as a *secondary* analysis. The primary evaluation—the baseline ladder—concerns caries only, because caries is the task for which we hold annotations on all five institutions.

TABLE I
Pipeline stages and model sizes. Parameter counts are read from the saved checkpoints, not from the architecture nominal.

Stage	Model	Parameters
1 Tooth detection	YOLOv11l-seg	27.62 M
2 FDI numbering	ResNet18 + 4 positional feats	11.20 M
3a Caries gate	ResNet18	11.19 M
3b Caries severity	ResNet18	11.19 M
4a Bone-loss gate	ResNet18	11.19 M
4b Bone-loss severity	ResNet18	11.19 M
Total		**83.56 M**
Caries path (stages 1–3)		*61.20 M*

IV. Data

A. Development corpora

Tooth detection and the bone-loss branch use the **Tufts Dental Database** [22] (1,000 OPGs) under a frozen 700/150/150 split. The caries branch and the primary evaluation use **DENTEX** [23]. FDI numbering trains on both, with a third public corpus used only for an ablation (Section VI-C).

All splits are **image-level**. Two crops from one radiograph are not independent observations, and a tooth-level split would place the same mouth in both training and test data.

Two labelling conventions were established by measurement rather than assumed, because a silent error in either would have trained a plausible but wrong model. Tufts numbering is left–right **mirrored** relative to Universal notation (Tufts "1" corresponds to FDI 28, not 18), confirmed three independent ways. The third corpus was checked the same way, using the two-corpus numbering head as an independent arbiter over 915 teeth: agreement was **80.3 %** reading its labels as written, against **1.7 %** reading them mirrored. Its labels are therefore standard FDI and are used as written.

B. Leakage control, and how it was established

The DENTEX `disease/` subset contains 250 annotated images. Our own models had already trained on some of them, so those must never be scored. Overlap was determined by **MD5 hashing every image and matching byte-identical files**, not by filename and not by assertion: 85 images are byte-identical to an image used in numbering-head training and 1 to an image used in detector training, giving **86 excluded** and **164 scored**. The record is stored as a manifest and re-checked at run time.

The image-blind constant is fitted on those excluded images and applied to the scored ones. Of the 86, **78 carry at least one caries annotation**, and those 78 are the fitting set. The two sets are disjoint by construction, and the fitting routine asserts the empty intersection before it runs.

We note the residual limitation honestly: DENTEX ships no patient identifiers, so we can establish image-level and

TABLE II
EVERY CORPUS USED, AND WHAT IT WAS USED FOR. "EXTERNAL" MEANS NO IMAGE FROM THAT CORPUS CONTRIBUTED TO TRAINING OR TUNING ANY MODEL IN THE PIPELINE. SPLITS ARE IMAGE-LEVEL THROUGHOUT.

Corpus	Role	Images	Annotation	Split used	External?	Fits baseline?
Tufts [22]	Detection, bone loss	1,000	tooth boxes; expert masks	700/150/150 frozen	no	no
DENTEX [23]	Caries; numbering	250 [a]	per-lesion polygons	164 scored / 86 excluded	no	78 of the 86
Kaggle dual-labelled	Numbering ablation	499	per-tooth boxes, FDI	18,372/1,961/1,994 crops	no	no
Ningxia	External test, locked	337 [b]	raster lesion masks	test1–test4, all scored	**yes**	no
CariesXrays [2]	External test	6,000 [c]	per-lesion boxes	random 1,500 scored	**yes**	no

[a] The DENTEX `disease/` subset, which is the portion carrying per-lesion annotation. [b] Caries-annotated subset; the periodontal subset of the same cohort is 736 images. [c] 13,783 annotated caries instances; we score a seed-fixed random 1,500 images carrying 3,421 lesions.

byte-level separation but not patient-level separation. If one patient contributed two radiographs under different filenames, we could not detect it.

C. External data

Four hospitals from an independent Chinese cohort (denoted test1–test4) serve as locked external test data: never trained on, never tuned against, and evaluated a bounded and recorded number of times. For caries this is 337 images carrying 1,013 annotated lesions; the periodontal subset of the same cohort is 736 images.

One property of that cohort must be stated because it invites an obvious question. The cohort ships its own train/test division, and **test2 is drawn from the same centre as that cohort's own training portion**, while test1, test3 and test4 are cross-centre. *We never trained on any Ningxia data*, so all four splits are equally external to our system; the label describes the source cohort's design, not our exposure. We report it because test2 is also the site where our pipeline scores highest, and a reader is entitled to know that the two facts coincide.

A second external cohort is the published **CariesXrays** benchmark [2]: 6,000 hospital-scale panoramic radiographs with 13,783 caries instances annotated by dental professionals, released with its own detection baselines. We score a seed-fixed random sample of 1,500 images carrying **3,421 lesions**—more than the development and first external cohorts combined—and verified it leakage-free against every corpus above, both byte-wise and perceptually, before use. Its licence is not stated by its authors; it is released publicly alongside the work cited here and used under that citation. Two properties bound what it can support and are reported rather than omitted: the corpus does not document whether its annotation is *exhaustive*, so a flagged tooth carrying no annotated lesion is not a confirmed false positive; and **22.1 %** of its lesions fall inside no detected tooth at all, which caps every tooth-level arm on it, ours included.

D. Crop geometry and pre-registration

Classifier inputs are expanded crops resized to 224×224. Expansion is fixed per branch and identical across corpora: symmetric $1.5\times$ for numbering, and $1.5\times$ width by $2.0\times$ height for bone loss, since teeth are roughly three times taller than wide and horizontal over-expansion bleeds into neighbouring teeth.

Operating points are fixed *a priori* and reported where they fall rather than tuned against evaluation data. One consequence is reported here rather than omitted: the detector's mask-quality bar was pre-registered at 0.65 and **missed**, at 0.638. The bar was not moved after the fact. The model was adopted for its bounding boxes alone—which improved from 0.782 to **0.810** mAP@0.5 on the same frozen split—by an explicit, dated decision, and the mask-dependent component it was intended to enable remains unwired.

V. EVALUATION PROTOCOL

A. The ladder

Four systems answer one question—*which teeth in this radiograph show caries?*—on the same 164 leakage-free DENTEX images, scored by one scorer per metric family. Each rung receives strictly less information than the one above.

1) **Pipeline**: the six stages of Section III.
2) **VLM, large**: Qwen2.5-VL-32B-Instruct [24], zero-shot; 33.45 B parameters as loaded.
3) **VLM, small**: Qwen2.5-VL-7B-Instruct, zero-shot; 8.29 B parameters as loaded.
4) **Constant**: a fixed set of FDI codes emitted for every image. **The radiograph is never read.**

B. How the constant is built

The constant is not hand-picked. We ran a greedy search over tooth codes on the 78 held-out fitting images, adding whichever code most improved F1 until no addition helped, and then froze the result. It selected $\{16, 17, 26, 27, 36, 37, 46, 47\}$—the first and second molars of all four quadrants.

Two properties are worth stating. First, the search never sees a scored image. Second, the frozen answer transfers *downwards*: it scores F1 0.392 on the images it was fitted on and 0.320 on the images it is scored on, which is what an honestly held-out fit should do.

We additionally report a variant fitted directly on the scored set. It is labelled an *oracle* throughout, is an upper bound on image-blind performance rather than a deployable baseline, and is not claimable.

C. Fairness

A rigged baseline is worse than no baseline. The VLMs are given the FDI numbering scheme in the prompt, told

explicitly that teeth may be absent, instructed to answer with codes alone, and parsed permissively, so that neither notation nor formatting is scored as clinical error. Few-shot exemplars, where used, are drawn strictly from the leakage-excluded images—which our own pipeline trained on—so the VLM receives free supervision our pipeline does not. Every affordance favours the opponent.

D. Metrics

We use two metric families, chosen because they fail differently. Each is defined in words before symbols, and each is reported with its limitation.

FDI-set F1 treats each image as a set-prediction problem: *of the tooth codes the expert marked carious, how many did the system name, and how many of the codes it named were not marked?* Precision is the fraction of predicted codes that are correct, recall the fraction of annotated codes that were found, and F1 their harmonic mean; all three are micro-averaged over teeth pooled across images. Its limitation is the subject of this paper: set membership over a fixed 32-code vocabulary can be answered partially from population statistics alone.

Lesion-localisation agreement asks, of every expert-annotated lesion, *was a tooth flagged that overlaps it in space?* It requires spatial correspondence, which set membership does not. On DENTEX, a lesion polygon is matched to a predicted-positive tooth box by greedy intersection-over-union at a threshold of 0.30 with a centroid-containment fallback. On the external cohort the matcher is centroid containment alone: those lesions are small focal blobs (median area 731 px) inside much larger tooth boxes ($\sim$25,000 px), and every true match scores an IoU of 0.015–0.124, below any usable threshold. **The two matchers are not interchangeable, and internal and external agreement figures are therefore not directly comparable to one another**; each is comparable across systems within its own cohort, which is what the comparison requires.

Lesion agreement has no negative class by construction—a deliberate property, since the external cohort contains no healthy controls—so it is a recall-shaped measure, and flagging more teeth can only raise it. We therefore always report it alongside the number of teeth flagged and the predicted-positive match rate, and we include an upper-bound arm that flags *every* detected tooth so the reader can see what saturating the metric costs.

Because these two families count different things, their denominators differ on the same images: the 164 scored radiographs carry **479 annotated caries lesion polygons** distributed over **477 distinct teeth**. The difference is exactly two teeth that each carry two separate annotated lesions. Both numbers are correct; they are different units, and each table states which it uses.

For the periodontal branch a third metric is unavoidable. Periodontal masks are not focal lesions but a single continuous band along the bone ridge, median area $\sim$365,680 px, whose centroid falls in the gap between teeth and inside no tooth box—so lesion-centroid matching returns near-zero by construction, independent of model quality. Periodontal findings are therefore scored by **affected-tooth sensitivity**: a tooth counts as affected if at least 50 % of its box lies inside the marked region, and sensitivity is the fraction of those the pipeline flagged.

E. Controls

For the VLMs, we present a **blank grey image** and a **random-noise image** under the identical prompt. Neither contains a tooth. For selective prediction, we **shuffle the confidences**, breaking the link between confidence and correctness while preserving both marginal distributions; a gain that survives shuffling comes from discarding items rather than from ranking them. For the localisation metric, the flag-every-tooth arm bounds the achievable range.

F. Statistical analysis

All confidence intervals are 95 % percentile intervals from a **cluster bootstrap resampled over images**, 10,000 resamples, seed 42. Differences between two systems evaluated on the same images are computed as paired differences within each resample.

Images, not teeth, are the resampling unit, and this is the choice to defend. The 936 teeth in the numbering evaluation come from 164 radiographs. Teeth within one image share a patient, a scanner, an exposure and an operator, and errors are correlated within an image—a mis-numbered quadrant mis-numbers several teeth at once. Treating 936 correlated teeth as 936 independent observations would produce intervals far too narrow. Where a metric's clustering differs, the number of clusters is reported with it.

Two quantities carry no interval and we say so rather than implying otherwise: external periodontal sensitivity, because affected-tooth counts were logged per split rather than per image and there is no cluster to resample; and detection mAP and the two classifier branches, whose per-item prediction dumps were not saved and which would need a re-evaluation pass to bootstrap.

VI. Results

A. The ladder on development data

Fig. 1 summarises the comparison; Table III gives the set-prediction result exactly.

The ordering is inverted with respect to the information each system receives. The best-performing entry never sees a radiograph, and the gap between the frozen constant and the test-set-fitted oracle is one percentage point—so almost all of the image-blind score is available without any access to the evaluation data.

A single metric behaving oddly is a metric artefact. We therefore repeated the comparison under lesion localisation, which demands spatial correspondence (Table IV).

The constant leads again, by **4.59 percentage points**, 95 % CI $[+0.60, +8.89]$. It does not achieve this by indiscriminate flagging: it flags 14 % more teeth than the pipeline at an

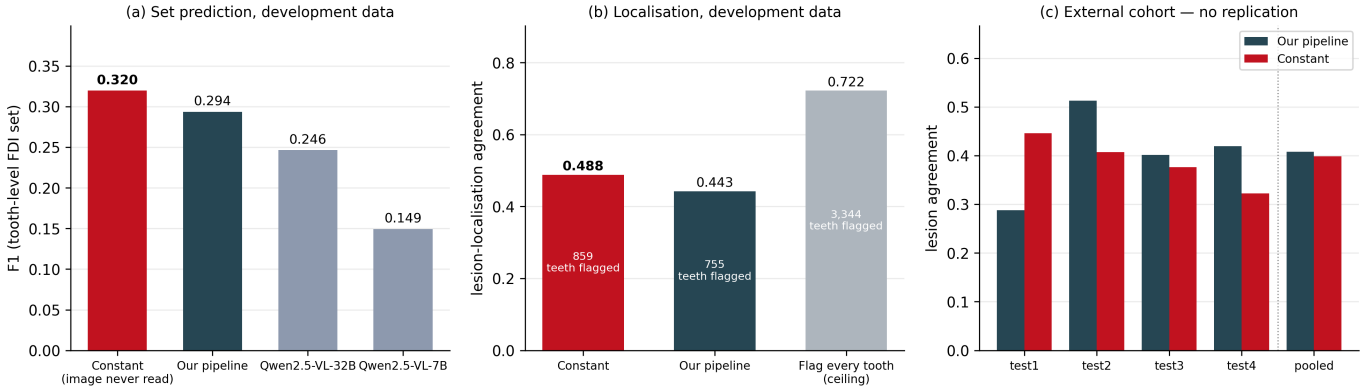


Fig. 1. The constant-baseline comparison. **(a)** Tooth-level FDI-set F1 on 164 leakage-free DENTEX development images (477 annotated carious teeth): the ordering is inverted with respect to the information each system receives, and the best-scoring entry never reads a radiograph. **(b)** A second, unrelated metric on the same images (479 annotated lesions), requiring the flagged tooth to overlap the lesion in space, returns the same ordering. Teeth flagged is annotated inside each bar because agreement is recall-shaped, and the flag-everything arm shows the metric's upper bound. **(c)** The external cohort (337 images, 1,013 lesions, four hospitals), where the development result does not replicate: the difference swings 26 points between sites and vanishes when pooled. Panels (a) and (b) are not comparable to (c): different cohorts and different lesion matchers (Section V). Every value is read from the saved evaluation artefacts at render time rather than transcribed.

TABLE III
TOOTH-LEVEL CARIES PREDICTION. **DATASET:** 164 LEAKAGE-FREE DENTEX IMAGES. **TASK:** NAME THE CARIOUS TEETH. **METRIC:** FDI-SET PRECISION/RECALL/F1, MICRO-AVERAGED. **UNIT:** TOOTH CODE; $n = 477$ ANNOTATED CARIOUS TEETH. ONE SCORER FOR ALL ARMS.

System	Precision	Recall	F1	Params
Constant (never reads image)	0.218	0.600	**0.320**	0
Our pipeline	0.235	0.390	0.294	61.2 M
Qwen2.5-VL-32B-Instruct, 0-shot	0.250	0.243	0.246	33.45 B
Qwen2.5-VL-7B-Instruct, 0-shot	0.248	0.107	0.149	8.29 B
Oracle constant (test-fitted)	*0.245*	*0.505*	*0.330*	*0*

TABLE IV
LESION LOCALISATION. **DATASET:** THE SAME 164 DENTEX IMAGES. **TASK:** FLAG A TOOTH OVERLAPPING EACH ANNOTATED LESION. **METRIC:** AGREEMENT (RECALL-SHAPED; NO NEGATIVE CLASS). **UNIT:** LESION; $n = 479$ ANNOTATED CARIES LESIONS. TEETH FLAGGED IS REPORTED BECAUSE AGREEMENT ALONE REWARDS FLAGGING MORE.

System	Agreement	Teeth flagged	Match rate
Constant	**0.488**	859	0.272
Our pipeline	0.443	755	0.281
All teeth (ceiling)	*0.722*	*3,344*	*0.103*

essentially identical match rate (0.272 versus 0.281). Two metric families designed to measure different things therefore agree, which moves the observation from an artefact of one scorer to a property of the data.

B. The ladder on external data

We then ran the same comparison on four hospitals the system had never seen (Table V).

The development-set result does not replicate. Pooled, the difference is -0.89 percentage points with 95 % CI $[-4.88, +3.01]$: the two systems are statistically indistinguishable, with the pipeline nominally ahead by an amount the interval cannot separate from zero.

TABLE V
EXTERNAL VALIDATION. **DATASET:** FOUR HOSPITALS, 337 IMAGES. **TASK:** FLAG A TOOTH CONTAINING EACH LESION CENTROID. **METRIC:** LESION AGREEMENT BY CENTROID CONTAINMENT—A DIFFERENT MATCHER FROM TABLE IV, FOR THE REASON GIVEN IN SECTION V. **UNIT:** LESION; $n = 1{,}013$.

Hospital	Centre	Lesions	Pipeline	Constant	Δ (pp)
test1	cross	316	0.288	**0.446**	+15.8
test2	same [a]	349	**0.513**	0.407	−10.6
test3	cross	162	0.401	0.377	−2.5
test4	cross	186	0.419	0.323	−9.7
Pooled	—	1,013	0.408	0.399	−0.89

[a] Same centre as the *source cohort's own* training portion. No Ningxia data was used to train or tune our system; all four splits are equally external to us (Section IV).

The per-site spread is the more useful observation. The difference swings from $+15.8$ points at test1 to -10.6 at test2—a 26-point range across four hospitals of a single cohort—and the pooled figure conceals this entirely. test1 is simultaneously the pipeline's worst site and the constant's best. We do not have an explanation for that, and we report it unexplained rather than omit it.

A second external cohort makes the point sharper still. We repeated the identical comparison, with the same matcher and the same frozen system, on the published CariesXrays benchmark [2]—a seed-fixed random 1,500 images carrying 3,421 lesions, more than the development and Ningxia cohorts combined (Table VI). There the constant does not merely tie: it wins by **25.2 percentage points**, 95 % CI $[+23.2, +27.3]$.

Two observations follow, and the second is the one that matters. First, the **brute-force explanation fails here**. On the development set the constant flagged only 14 % more teeth at

TABLE VI
THE SAME COMPARISON ACROSS THREE COHORTS. **METRIC:** LESION AGREEMENT BY CENTROID CONTAINMENT FOR THE TWO EXTERNAL ROWS; THE DEVELOPMENT ROW USES THE MATCHER OF TABLE IV AND IS NOT COMPARABLE IN ABSOLUTE VALUE. **UNIT:** LESION. INTERVALS ARE PAIRED CLUSTER BOOTSTRAPS OVER IMAGES.

Cohort	Lesions	Pipeline	Constant	Δ (pp), 95% CI
DENTEX (development)	479	0.443	**0.488**	$+4.6$ $[+0.6, +8.9]$
Ningxia (4 hospitals)	1,013	0.408	0.399	-0.9 $[-4.9, +3.0]$
CariesXrays	3,421	0.309	**0.561**	$\mathbf{+25.2}$ $\mathbf{[+23.2, +27.3]}$

an equal hit rate, so volume did not account for its lead. On CariesXrays it flags 64 % more teeth *and* attains the higher hit rate (0.206 against 0.190): it is ahead on both axes.

Second, **the size of the gap is a property of the cohort, not of the systems**. Assigning every CariesXrays lesion to the tooth containing it, **72.0 %** sit on molars and **60.2 %** on precisely the eight codes the constant emits—against a structural ceiling near 0.60 for that constant. Where caries concentrates in the molars, an anatomical prior is close to optimal by construction.

Taken together the three cohorts do not support the claim that our pipeline outperforms an image-blind prior on these metrics, and they place the margin anywhere between a tie and 25 points depending only on which cohort is scored. The swing across a single cohort's four sites is 26 points; the swing across cohorts is 25. **A tooth-level agreement figure reported on one cohort therefore carries almost no information about a system's perceptual ability**, which is a stronger and better-supported statement than the development set alone would license.

C. What the constant cannot do

A constant answers a set-membership question. It cannot perform tasks whose output is not a fixed list, and the pipeline's results on those tasks are correspondingly meaningful.

Tooth numbering. End-to-end FDI-exact accuracy is 58.3 % (546/936 teeth), 95 % CI $[53.9, 62.8]$. This figure is **detection-bound**, and we report it as two numbers rather than one: 29.4 % of ground-truth teeth are never detected, and of the 661 teeth Stage 1 does find, Stage 2 numbers **82.6 %** correctly. It is the second figure that compares to the pre-cropped accuracies in Section II; the first is what an end-to-end system delivers.

Residual errors are largely benign: of 115 incorrect codes, 89 (77.4 %) name an adjacent tooth within the correct quadrant, and only 16 of 661 detected teeth (2.4 %) receive an error at the quadrant level—naming a tooth in the wrong corner of the mouth. A constant cannot attempt this task at all.

Calibration. *Calibration* asks whether a model's confidence behaves like a probability: of the teeth it calls carious with 80 % confidence, are 80 % actually carious? **Expected calibration error** (ECE) is the average gap between stated confidence and observed frequency across confidence bins; 0 is perfect.

The caries gate's confidences are not probabilities. Its ECE is **0.136** and every reliability bin lies below the diagonal—it is systematically overconfident. Temperature scaling [25], the standard first remedy, made calibration *worse* (0.152), indicating the miscalibration is structural rather than a single scale factor. Isotonic regression [26] reduced ECE to **0.043**.

That improvement is not free, and we report its cost. Isotonic regression is only weakly monotonic: on this model it collapses the test scores to 46 distinct values and average precision falls from **0.550 to 0.500**. Applied at the decision threshold it would also change **155 of 863 verdicts (18 %)**. We therefore apply it to the *reported confidence only*, leaving each verdict as the raw `argmax`, so that every published caries number in this paper is the one that was measured. This is a deployment choice with a stated price, not a property of the method—and it is the opposite of temperature scaling, which cannot change a verdict but did not fix the calibration.

One number should accompany all of this: the gate's average precision is **0.550** $[0.445, 0.660]$ against a prevalence of 0.147, i.e. 3.7$\times$ chance. What fails is calibration and the operating point, not the model's ability to rank teeth.

Selective prediction. A system may be allowed to *decline*—to answer "I am not confident enough on this tooth" instead of guessing. *Coverage* is the fraction of cases it still answers; accuracy is then measured on those alone. This is only useful if the model's confidence actually identifies its own errors.

Declining the least-confident quarter of cases raises numbering accuracy from 0.826 to **0.885** ($+5.91$ pp, CI $[+3.88, +7.68]$, 151 image clusters) and caries-gate accuracy from 0.797 to **0.940** ($+14.25$ pp, CI $[+11.32, +16.63]$, 34 image clusters). Under the shuffle control both gains collapse to $+0.03$ and $+0.00$ percentage points, establishing that the improvement comes from the confidence ranking and not from discarding items.

The pre-registered threshold of 0.30 lands at 90.8 % coverage and 0.847 accuracy, within 0.2 points of the empirical 90 %-coverage operating point (0.849)—a value chosen before these data were seen.

One caveat must accompany any coverage figure here: coverage is measured over *detected* teeth. Retaining 75 % of detections corresponds to 75 % of 70.6 %, or **53.0 % of the teeth actually present in the mouth**.

D. Vision–language model behaviour

Few-shot prompting, with exemplars drawn strictly from leakage-excluded images, degrades the smaller model monotonically: F1 falls from 0.149 zero-shot to 0.066 at two shots and 0.034 at four. The mechanism is abstention rather than error—empty answers rise from 56 of 164 images to 123 of 164—so exact-set match simultaneously *rises*, because 16 images contain no caries and an empty answer scores an exact match on each. Reported alone, that column would be true and thoroughly misleading.

The content-free controls are more direct, and the two models behave differently enough that they must be reported separately.

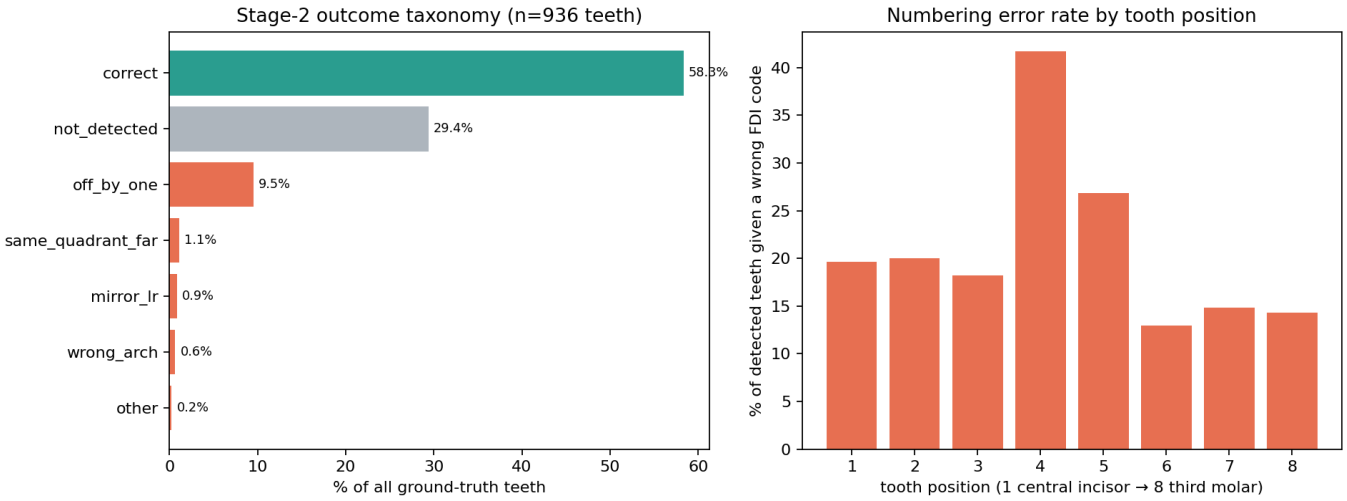


Fig. 2. Where the numbering stage loses teeth, over 936 ground-truth teeth in 164 images. The largest single category is teeth the detector never proposes, which no numbering model can recover; among the codes that are wrong, most are off-by-one within the correct quadrant.

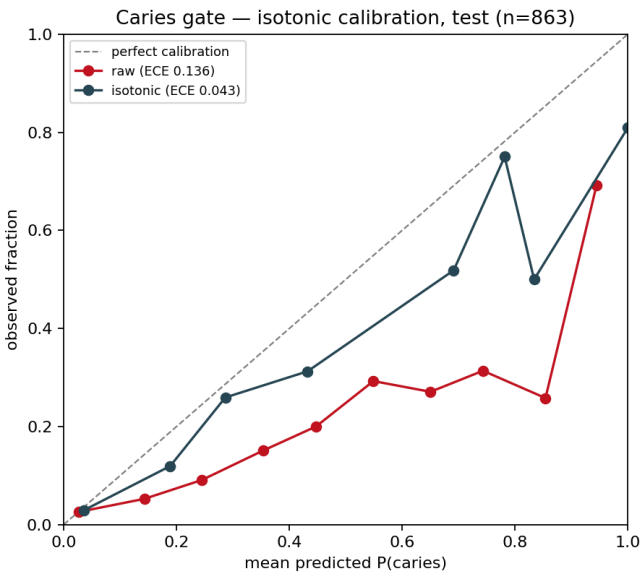


Fig. 3. Reliability of the caries gate before and after isotonic regression, on 863 held-out test crops. The uncorrected model is systematically overconfident: at a stated confidence of 0.85 the observed frequency of caries is 0.26.

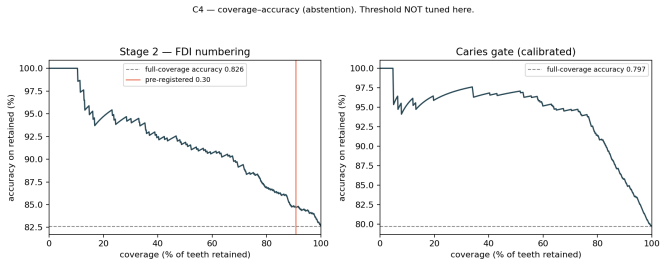


Fig. 4. Coverage–accuracy for tooth numbering (left) and the caries gate (right). Accuracy on retained predictions rises as the least-confident cases are declined. The vertical line marks the pre-registered threshold of 0.30, located on the curve rather than tuned against it.

- **Qwen2.5-VL-7B-Instruct** answers `26,36` to a blank grey image and `26,36` to random noise—identical to its answer for a real radiograph.
- **Qwen2.5-VL-32B-Instruct** answers `NONE` to the blank image, which is correct behaviour. Shown random noise, it answers `16, 17, 26, 27, 36, 37, 46, 47`—*exactly* the eight-molar set our greedy search recovered from the data, and identical to its answer for a real radiograph.

The larger model therefore detects the absence of an image but not the absence of anatomy, and what it emits in that condition is the population prior itself. Whatever these outputs measure on a noise input, it is not perception of the image in front of them. We claim this for these two models under this prompt, and no further.

The efficiency comparison survives all of the above: 33.45 B parameters and 70.1 GB of accelerator memory are outscored by 61.2 M parameters running on a CPU—a factor of roughly 550 in parameter count, in the wrong direction for the larger model.

E. Where the pipeline fails: the periodontal branch

The bone-loss branch reaches **0.923 macro-F1** in domain on 1,558 Tufts validation crops at an untouched 4 % prevalence (sensitivity 83.05 %, specificity 99.53 %). Outside that domain it is close to inert: it fires on **0.7 %** of DENTEX teeth and reaches **1.4–5.6 %** affected-tooth sensitivity across the four external hospitals. Two independent external datasets, with different ground-truth formats and different metrics, agree.

Two caveats belong with the in-domain figure, and both make the collapse less surprising rather than more. First, 0.923 is the *maximum of a noisy selection curve*: the diseased-class

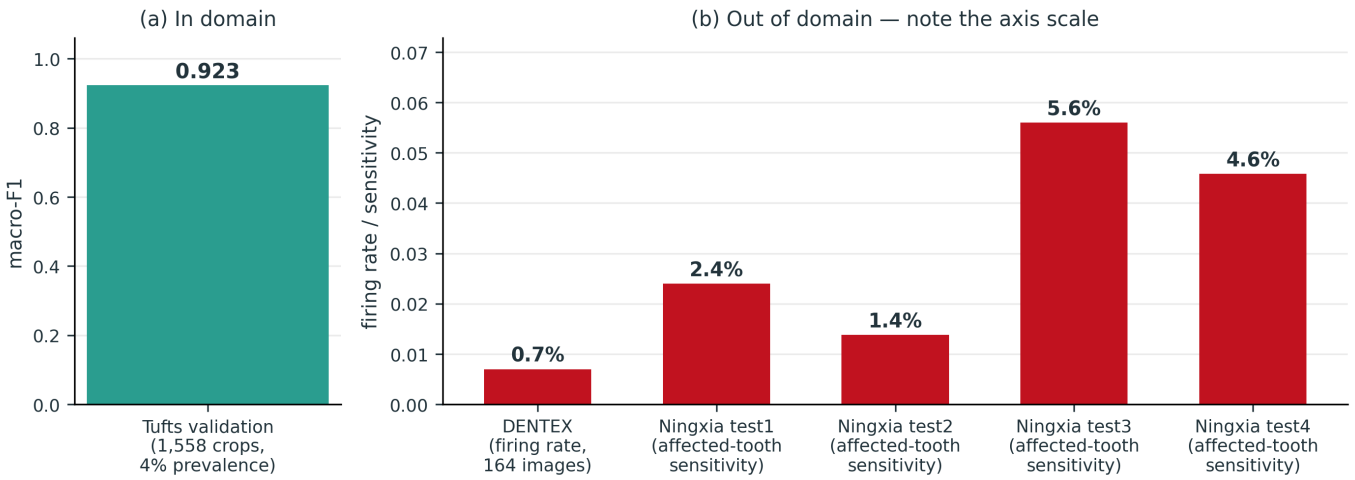


Fig. 5. The bone-loss branch under domain shift. **(a)** In-domain macro-F1 of 0.923 on 1,558 Tufts validation crops at an untouched 4 % prevalence. **(b)** The same frozen model outside its training domain, *on a different vertical scale*: it fires on 0.7 % of DENTEX teeth and reaches 1.4–5.6 % affected-tooth sensitivity across the four external hospitals. The metrics necessarily differ between cohorts, which is why they are not plotted on a shared axis; the collapse does not depend on that choice. Reported as a measured negative result rather than as a component.

validation F1 oscillates between roughly 0.40 and 0.85 across training epochs over only 59 validation positives, so selecting the best of 32 epochs is optimistically biased and a repeated-split estimate would very likely sit lower. Second, **the figure is not currently reproducible.** Half the validation crops were built with a coordinate jitter whose random offsets were never saved, and the source radiographs the crops were cut from are not on the machine that would rebuild them; a reconstruction attempt was rejected by its own geometry check rather than published.

We localised the cause. A detector change that moved numbering by $+12.0$ points and caries agreement by $+7.7$ moved bone loss by *nothing*—the branch flagged 168 affected teeth before the swap and 165 after, while the denominator grew 9 %. That falsifies the hypothesis that a shared upstream cause explains all three failures, and places this one inside the Stage-4a classifier rather than in the pipeline around it.

We did not attempt a fix, and the reason is a data fact rather than a choice: the lever that repaired the numbering stage's cross-domain behaviour was multi-corpus training, and **DENTEX carries no bone-loss labels**. There is no second annotated periodontal corpus in this project, so that lever does not exist here.

VII. Discussion

A. Main finding

On our development data an image-blind constant outscored a purpose-built six-stage pipeline and two vision–language models under two unrelated metrics. On four external hospitals the same comparison produced no separation. The finding is therefore not that the constant wins—it does not, reliably—but that **these metrics cannot tell the two apart**, which is a stronger and more troubling result than either outcome alone.

B. Why the floor is high, and why that is not a criticism of any model

Caries in these cohorts is anatomically concentrated: molars carry most of the disease. A tooth-level decision is a set-membership question over a fixed vocabulary of 32 codes, and set membership inherits that concentration directly. A fixed answer naming the eight first and second molars is not a degenerate output—it is a compact statement of dental epidemiology, and it is right for the same reason a clinician's prior is right.

The consequence is that the headroom above such an answer is small, and small *independently of model quality*. No amount of architectural work raises a ceiling imposed by the pairing of task and metric. A system distinguishes itself here only by being right about *this patient* in a way the population prior is not, and tooth-level set membership is a poor instrument for detecting that difference.

That two unrelated metrics agree on development data is what makes this a property rather than an anecdote. Set-membership F1 and lesion-localisation agreement were constructed to measure different things, and the second was adopted precisely because it demands spatial correspondence.

C. Why the external disagreement changes the interpretation

We expected the external cohort to confirm the development result. It did not. Pooled, the two systems are indistinguishable; per site, the difference swings through 26 points.

Two readings remain open and we cannot separate them with the data we have. Either the development result is partly a property of DENTEX's particular caries distribution, or the external metric—which must use centroid containment for the geometric reason given in Section V—has different sensitivity to the same underlying effect. We report the ambiguity rather than resolve it by assertion.

What the disagreement does establish is firmer than what it leaves open: at this variance, *a single external number carries very little information*. Our own pooled 0.408 is, in isolation, as uninterpretable as any other pooled figure in this literature, including the ones we cite. One hospital in our cohort would have supported the claim that the constant wins decisively; another would have supported the opposite. Reporting either alone would have been defensible, publishable, and wrong.

D. What the system can still demonstrate

A constant cannot name a tooth, cannot express a calibrated confidence, cannot decline informatively, and has no behaviour under domain shift to measure. We propose these as the axes on which OPG systems should be compared, with our own figures as an illustration of what reporting them looks like:

1) **Identification**—can it say *which* tooth? Here 82.6 % of detected teeth, with the 70.6 % detection bound stated alongside.
2) **Calibration**—do its confidences behave like probabilities? ECE $0.136 \rightarrow 0.043$, with the failed temperature-scaling attempt reported because it diagnoses the miscalibration as structural, and with the $0.550 \rightarrow 0.500$ average-precision cost stated.
3) **Selective prediction**—does declining improve accuracy, and does the gain survive a shuffle control? $+14.25$ pp, and yes.
4) **Domain shift**—measured on held-out institutions and reported per site, including when the answer is that a branch does not transfer.

A constant scores nothing on any of these. That asymmetry, rather than a caries F1, is what the system has to show.

E. Anticipated objections

"Your pipeline is simply weak." Possibly, but that does not explain the result. Two general-purpose models at 8.29 B and 33.45 B parameters were scored by the identical scorer and placed *below* our pipeline, and all three below the constant. An explanation resting on our system's quality would have to account for three systems of very different design failing to separate from the same prior.

"The constant was fitted, so it is not uninformed." It is fitted—on 78 held-out images, byte-disjoint from every scored image—and then frozen. What it learns is a population statistic: which teeth tend to be carious, not anything about the radiograph in front of it. That is exactly the quantity a diagnostic claim must exceed. Its score drops from 0.392 on the fitting images to 0.320 on the scored ones, in the direction an honest held-out fit should.

"The metrics were poorly chosen." The first is the metric this literature predominantly reports; we did not select it to make a point. The second we introduced specifically to escape the first, and it produced the same ordering. We also report the degenerate flag-everything arm, which reaches 0.722 agreement, so the reader can judge the metric's headroom directly.

"A molar-prior system would still be clinically useful." It might well be, and we do not dispute it. But a tool that encodes epidemiology is a different object from a tool that reads a radiograph, and only the second can find the lesion the prior does not expect. That distinction is invisible to the metric and central to the clinic.

F. What we propose should be reported

An image-blind baseline row. One line of code, no compute, alongside any tooth-level caries metric on this class of data. Without it a mid-range number cannot be interpreted—our four systems span 0.149 to 0.320 with the *least* informed at the top.

Per-institution results. Pooled external figures conceal exactly the variation that external validation exists to expose.

Metrics with demonstrated headroom. If a degenerate arm reaches most of the achievable range, the metric cannot support a perception claim regardless of what a system scores on it.

None of this invalidates prior work. It narrows one specific inference: the claim *this system perceives caries* is not established by a tooth-level number alone.

G. Clinical interpretation

This study does not establish clinical usefulness, and nothing here should be read as a claim about patient outcomes. We did not compare any system against a dentist, did not evaluate prospectively, and did not measure whether a flagged tooth changes what a clinician does. What the study does support is a narrower statement about how such systems should be assessed before they reach a clinic: that a tooth-level agreement score, reported without a floor and without external per-site results, is not sufficient evidence that a screening aid is reading the image it is shown.

H. On reporting results that do not favour the authors

Two findings here are unfavourable to our own system: the constant comparison and a periodontal branch that does not generalise. We report both. A screening tool that fails silently is more dangerous than one that fails visibly, and the alternative—selecting whichever metric flatters the system—is precisely the practice this paper argues against. We would have no standing to recommend image-blind baselines to others while declining to survive one ourselves.

VIII. Limitations

Each item states what it prevents us from concluding.

1) **No clinician comparison.** We do not know how any system here compares against a dentist on these images. We therefore cannot claim clinical utility, non-inferiority, or that the constant would mislead a clinician less than the pipeline. This is the single largest gap.
2) **Two caries cohorts disagreeing in direction.** The constant leads on development data and ties externally. We cannot claim the constant wins in general; the supported claim is non-separability.

3) **Internal and external caries figures are not comparable to each other.** They use different lesion matchers, forced by a 200-fold difference in lesion-to-tooth area. No internal-versus-external degradation figure should be read off Tables IV and V.
4) **Tooth-level is our finest localisation.** The localisation metric is a tooth-level correspondence check, not a pixel-level segmentation assessment. We cannot claim the system localises *within* a tooth.
5) **The external cohort lacks healthy controls.** Specificity, AUC and false-positive rate are not computable there; we report agreement only, which cannot be penalised for over-flagging.
6) **Unmatched predicted-positive teeth are not confirmed false positives.** DENTEX annotates only teeth carrying some finding—roughly 6 of about 30 per image—so most teeth are unannotated rather than verified healthy. The match rate is reported as a diagnostic, never as precision.
7) **One VLM family at two scales.** A domain-adapted medical VLM was not accessible. We cannot generalise the control result to VLMs as a class, only to these two models under this prompt.
8) **Detection bounds Stage 2.** At 0.810 box mAP@0.5, 29.4 % of teeth are never presented to the numbering head, and all coverage figures are fractions of the 70.6 % detected.
9) **The in-domain bone-loss figure is a selected maximum and is not currently reproducible** (Section VI). It should be read as an upper bound on that branch's in-domain performance, not an estimate.
10) **Bone-loss labels are weak supervision**, derived by intersecting a per-image expert mask with per-tooth boxes, from a single institution and one annotation protocol, with only 711 real positive training teeth.
11) **Patient-level separation cannot be verified on DENTEX**, which ships no patient identifiers. We establish byte-level and image-level separation only.
12) **All data is retrospective and public or archival.** There is no prospective evaluation, and prevalence in these cohorts need not match any clinic's.

IX. Conclusion

We evaluated an automated screening pipeline for dental caries and periodontal bone loss on panoramic radiographs, and measured what its scores mean by comparing it against progressively less informed systems under one scorer.

On development data, a fixed list of eight molar codes—emitted for every radiograph, fitted on held-out images and frozen—outscored a six-stage pipeline and two vision–language models under two unrelated metrics. A tooth-level score in this range therefore cannot on its own establish that a system is using image information. Across four external hospitals the same comparison produced no separation at all, with a 26-point swing between sites that the pooled figure erases; a single pooled external number carries correspondingly little information. Both language models returned confident, correctly formatted, anatomically specific answers to an image containing pure noise, and the larger one's answer in that condition was precisely the population prior. The periodontal branch scored 0.923 macro-F1 in domain and did not transfer.

Four capabilities distinguished the system from the prior where the caries score could not: identifying which tooth, producing confidences that behave like probabilities, declining informatively with a gain that survives a shuffle control, and being measurable under domain shift. These cost little to report.

We therefore recommend three things of the next paper reporting a tooth-level caries result: publish an image-blind baseline beside it, report external results per institution rather than pooled, and state the headroom of the metric being used. We do not conclude that automated caries screening does not work. We conclude that this measurement cannot tell us whether it does.

Reproducibility

Every figure is rendered from the saved evaluation artefacts at build time rather than transcribed. All model checkpoints are SHA-256 pinned, and each evaluation artefact records the hashes of the weights that produced it. The frozen constant reported in Table III was independently re-derived from the 78 fitting images during manuscript preparation and reproduced exactly. The supporting test suite comprises 122 tests, all passing.

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